Vocational Training Development Institute

SCHOOL OF APPLIED TECHNOLOGY

INFORMATION AND COMMUNICATION TECHNOLOGY

DYNAMIC WEBSITES & APPLICATIONS

Module No. 1

Assignment

Due: - February **Total marks:** Module 1 - **60 marks**

- *Date*: February 26, 2025 *Lecturer*: Doron Williams

Instructions:-This is a group assignment.

Complete each question in separate files using JavaScript for your answers

1. Display the sum of two numbers inside a paragraph

- Create two variables, `x` and `y`, and store the values `5` and `10` in them.
- Add these two numbers together.
- Display the result inside the `<p>` element with the ID `demo`.

```
html
CopyEdit
<p id="demo"></p>
```

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2. Prompt the user for their first name, last name, and age

- Declare three variables (`firstName`, `lastName`, and `age`) on a single line.
- Use `prompt()` to ask the user for their first name, last name, and age.
- Display the collected values in an alert box.

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3. Add a red border to all `<textarea>` and `<p>` elements when a button is clicked

- Find all `<textarea>` and `<p>` elements in the document.
- Set their border color to red.
- Make sure this happens when the button is clicked.

```
html
<body>
  <textarea>HTML & CSS gives us structure and style</textarea>
  <textarea>JavaScript brings our webpage to life!!</textarea>
  <p>jQuery</p>
  <p>JavaScript</p>
  <button id="button1">Click to check the effect</button>
</body>
```

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4. Create a paragraph and append text using Template Literals

- Create a new paragraph (`<p>` element).
- Use Template Literals (backticks ` `) to set the paragraph's text.
- Append the paragraph to the body of the page.

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5. Change the font color and background color of a specific paragraph when clicked

- Add a click event to the paragraph with the ID `targetparagraph`.
- When clicked, change its text color and background color.

```
html
<p>PHP</p>
<p>Java</p>
<p id="targetparagraph">Python</p>
```

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6. Change the background color of `<div>` elements that contain "tutorial"

- Add a class name to the `<div>` elements that contain the word "tutorial".
- Use JavaScript to find these elements and change their background color to yellow when a button is clicked.

```
html
<body>
  <div class="tutorial">
    <p>PHP</p>
  </div>
  <div>
    <p>Java</p>
  </div>
  <div class="tutorial">
    <p>Python</p>
  </div>
  <button id="btn1">Click to see the effect</button>
</body>
```

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7. Display a message based on the user's selection from a dropdown

- Create a dropdown (`<select>` element) with a few options.
- When the user selects an option, display a message using an `if..else` or `switch` statement.

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8. Change an image's `src` attribute

- Find an image element with an `id` of `"image"`.
- Change its `src` attribute to `"mountain.jpg"` when the page loads.

```
html
<img id="image" src="smiley.gif" alt="Picture of smile emoji">
```

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9. Retrieve and display the style properties of the document

- Write a function that finds the `<style>` element in the document.
- Extract its contents and display them inside the `<p>` element with the ID `demo`.

```
html
<p>Click the button to display the style properties of this document.</p>
<button onclick="myFunction()">Try it</button>
<p id="demo"></p>
```

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10. Change the text color of any paragraph when the mouse hovers over it

- Find all `<p>` elements in the document.
- Change the text color to red when the mouse is over any paragraph.

```
html
<body>
  <p>jQuery Exercises, Practice and Solution.</p>
  <p>Ajax Tutorial</p>
</body>
```

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11. Change font weight and color when the mouse enters and leaves

- When the mouse enters a paragraph, change its font to bold and color to blue.
- When the mouse leaves, reset it to normal and black.

```
html
<body>
  <p>Move the mouse over this and next paragraph.</p>
```

```
  <p>Web development tutorials</p>
</body>
```

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12. Remove a paragraph when a button is clicked

- Define a function that removes the paragraph with the ID `"demo"` when the button is clicked.

```
html
<p id="demo">Click the button, and this paragraph will be removed from the
DOM.</p>
<button onclick="myFunction()">Remove paragraph</button>
```

---

13. Display the last item of an array inside a paragraph

- Create an array with some fruit names.
- Get the last item from the array.
- Display it inside the `<p>` element with the ID `"demo"`.

```
html
<p>JavaScript array elements are accessed using numeric indexes (starting
from 0).</p>
<p id="demo"></p>
```

1.